
Fisher-Preconditioned Contrastive Alignment for Teacher-Free Test-Time Model Merging

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Abstract

Test-time model merging (TTMM) has emerged as a powerful, backpropagation-free alternative to traditional test-time adaptation (TTA), dynamically interpolating the parameters of specialized expert models to handle non-stationary test streams during inference. However, existing teacher-free TTMM frameworks rely on un-regularized prediction entropy minimization, which ignores the highly heterogeneous parameter sensitivities across neural network layers, resulting in representation collapse and catastrophic parameter drift. To overcome this fundamental limitation, we propose Fisher-Preconditioned Contrastive Alignment (FP-CA), a novel, completely self-supervised, teacher-free TTMM framework. FP-CA integrates three core contributions: (1) Prototype-driven Dynamic Routing (PD-Routing) which leverages pre-stored lightweight class prototypes of the expert models to perform highly accurate, unsupervised task detection and dynamically initialize coefficients, (2) Confidence-Masked Contrastive Alignment which optimizes the InfoNCE contrastive loss between online features and active prototypes to eliminate confirmation bias, and (3) Layer-wise Fisher sensitivity preconditioning which scales layer-specific learning rates dynamically using pre-computed diagonal Fisher Information to damp updates in highly sensitive layers and accelerate them in robust layers. Extensive evaluations on a multi-task vision benchmark under severe environmental corruptions demonstrate that FP-CA completely pre-

vents representation collapse, outperforming standard TTMM baselines across both rapid alternating and block-sequential streams.

1. Introduction

1.1. Background and Motivation

In real-world applications, deep learning models are frequently deployed in highly dynamic and non-stationary environments where the target data distribution drifts continuously from the original training distribution (Wang et al., 2021; Liang et al., 2020; Song et al., 2023; Du et al., 2024; Lin et al., 2024; Ruan & Tang, 2024). This phenomenon, known as out-of-distribution (OOD) shift or domain shift, can degrade model performance dramatically (Döbler et al., 2022; Zhou et al., 2024). While fully adapting model parameters on-the-fly at inference time is a natural solution, standard Test-Time Adaptation (TTA) approaches (e.g., TENT (Wang et al., 2021)) incur prohibitive computational costs and are highly prone to representation collapse, catastrophic forgetting, or parameter drift under continuous, long-term non-stationary streams (Wang et al., 2022; Zhao et al., 2023; Niu et al., 2023).

To address these limitations, model merging has emerged as an elegant, parameter-space paradigm (Yadav et al., 2023; Ilharco et al., 2022). Instead of training models from scratch on multiple datasets, model merging integrates the capabilities of independently fine-tuned “expert” models sharing the same loss basin into a single multi-task unified network. Recently, Test-Time Model Merging (TTMM) has been proposed to combine TTA and model merging (Yang et al., 2024; Anonymous, 2026b). Instead of modifying the millions of parameters of the base model, TTMM dynamically solves for a set of low-dimensional merging coefficients (e.g., layer-wise or global task-routing weights) on-the-fly, combining expert parameters in response to the active test distribution.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1.2. Challenges in Test-Time Model Merging

Despite its parameter efficiency, teacher-free TTMM faces two severe, unresolved challenges:

1. **Softmax Saturation and Momentum Lag:** Under sequential streams, where a single task persists for many batches before transitioning to another, optimization objectives like entropy minimization quickly push the merging coefficients to extreme values. When a task transition occurs, the accumulated optimizer momentum and softmax saturation prevent the model from adapting quickly, leading to massive lag and catastrophic misclassifications on the new task.
2. **Uniform Learning Rate Collapse:** Standard TTMM updates all layers' merging coefficients using a single uniform learning rate (Yang et al., 2024). However, deep neural networks exhibit highly heterogeneous parameter sensitivities (Anonymous, 2026a). Sensitive layers (such as early convolutional feature extractors and final classification heads) collapse and drift rapidly under unconstrained optimization, while robust, highly-expressive representation layers adapt too slowly, leading to sub-optimal solutions and performance degradation.

To bypass these trade-offs, some recent approaches employ unmerged expert models in memory as on-line teachers to guide adaptation (Anonymous, 2026c). However, storing multiple full expert models in GPU memory incurs a massive, often prohibitive memory footprint, violating the core motivation of lightweight inference-time adaptation.

1.3. Proposed Solution: FP-CA

To overcome these fundamental challenges, we propose Fisher-Preconditioned Contrastive Alignment (FP-CA), a mathematically principled, teacher-free, and highly robust framework designed to stabilize and accelerate test-time model merging. FP-CA consists of three tightly coupled components:

- **Prototype-Driven Dynamic Routing (PD-Routing):** We pre-extract and store extremely lightweight class prototypes from a small calibration set. For each incoming test batch, we perform a fast forward anchor pass to estimate task affinities, allowing us to initialize or reset coefficients instantly and clear optimizer momentum at task boundaries, completely bypassing momentum lag.

- **Confidence-Masked Contrastive Alignment:** To align representations without the memory overhead of teacher models, we optimize a Confidence-Masked contrastive InfoNCE loss against active prototypes, filtering out low-confidence samples to eliminate confirmation bias under noise.
- **Layer-wise Fisher Sensitivity Preconditioning:** We utilize pre-computed diagonal Fisher Information as a structural sensitivity prior to scale layer-wise learning rates dynamically. This scales down updates in highly sensitive layers to prevent representation collapse, while accelerating adaptation in robust layers.

Through extensive evaluations on a multi-task vision stream benchmark (MNIST, FashionMNIST, and KMNIST) under clean and highly corrupted settings, we demonstrate that FP-CA achieves a decisive victory. Under severe Gaussian Noise, FP-CA boosts average accuracy from 29.54% (standard AdaMerging) to 73.46%, completely preventing representation collapse and tracking task boundaries with exceptional stability.

1.4. Summary of Contributions

In summary, our key contributions are:

- We analyze the vulnerability of teacher-free test-time model merging to uniform learning rate collapse and softmax saturation under non-stationary streams.
- We introduce FP-CA, a novel, completely self-supervised, teacher-free TTMM framework that integrates prototype-driven routing, confidence-masked contrastive alignment, and layer-wise Fisher preconditioning.
- We prove empirically that scaling layer-specific learning rates based on diagonal Fisher sensitivities prevents representational drift and stabilizes coefficient trajectories.
- We demonstrate that FP-CA establishes a new state-of-the-art across alternating and block-sequential test streams under clean and highly corrupted environments, outperforming standard baselines by a large margin.

2. Related Work

2.1. Fully Test-Time Adaptation (TTA)

Test-Time Adaptation aims to adapt a pre-trained source model to unlabelled target domains during in-

ference (Liang et al., 2020; Wang et al., 2021; Karmanov et al., 2024; Boudiaf et al., 2022; Mounsaveng et al., 2023; Gan et al., 2022; Press et al., 2023). TENT (Wang et al., 2021) pioneered this by minimizing prediction entropy to update batch normalization parameters. SHOT (Liang et al., 2020) freezes the classifier head and maximizes mutual information while employing pseudo-labeling. CoTTA (Wang et al., 2022) and Delta (Zhao et al., 2023) focus on continual, long-term TTA, introducing mean-teacher weight averaging and degradation-free objectives to prevent error accumulation. Other variants like EATA (Niu et al., 2023), NOTE (Gong et al., 2022), and RoTTA (Yuan et al., 2023) introduce sample selection, memory buffers, or robust contrastive objectives (e.g., (Chen et al., 2022; Lee et al., 2024b; Zhou et al., 2025; Lee et al., 2024a; Zhang et al., 2024; Ye et al., 2025; Cui et al., 2025; Guo & Jin, 2025)). However, these methods require full backpropagation through millions of parameters, which is computationally expensive and prone to representation collapse.

2.2. Model Merging and Weight Averaging

Model merging is an emerging paradigm to integrate multiple specialized models fine-tuned from a shared pre-trained base model into a single multi-task unified model without joint training (Yadav et al., 2023; Ilharco et al., 2022; Choi et al., 2024; Bhardwaj et al., 2024; Tang et al., 2025; Li et al., 2025). Task Arithmetic (Ilharco et al., 2022) demonstrates that linear operations on task vectors (fine-tuned weights minus base weights) can edit model capabilities. TIES-Merging (Yadav et al., 2023) resolves parameter-level interference by trimming redundant parameters and resolving sign conflicts. DARE (Choi et al., 2024) applies random pruning to task vectors to keep only significant parameters before merging. Model Soups (Gu et al., 2023; Jia et al., 2023; Guo et al., 2022) and related weight-averaging methods (e.g., (Sanyal et al., 2023; Kaddour, 2022; Eeck & hamme, 2022; Lu et al., 2022; Athiwaratkun et al., 2018; Cha et al., 2021; Nikishin et al., 2018; Wan et al., 2021; Kleiman et al., 2025; Xie et al., 2025; Fassold, 2023)) interpolate checkpoints to improve generalization. Although highly efficient, traditional model merging relies on static, manually searched merging coefficients, which cannot adapt dynamically to non-stationary online environments.

2.3. Parameter Sensitivity and Preconditioning

Analyzing parameter sensitivity is a cornerstone of deep learning optimization and continual learning. Elastic Weight Consolidation (EWC) (Ales et al., 2017; 2016) uses diagonal Fisher Information to iden-

tify critical parameters and penalize their modification to prevent catastrophic forgetting. Natural Gradient Descent and preconditioning methods scale parameter updates using curvature information to accelerate optimization. In the context of model merging, Fisher Information has been utilized to perform weighted parameter averaging (e.g., Fisher-Merging (Anonymous, 2026a)), but static formulations cannot handle real-time distribution shifts during inference.

2.4. Test-Time Model Merging (TTMM)

To bridge the gap between static model merging and dynamic real-time adaptation, Test-Time Model Merging has been proposed (Yang et al., 2024; Anonymous, 2026b; Motyka et al., 2025; Li et al., 2024; Corbeil et al., 2025; Gargiulo et al., 2024; Panariello et al., 2025; Kuo et al., 2025; Du et al., 2025). AdaMerging (Yang et al., 2024) optimizes layer-wise merging coefficients at test time by minimizing entropy on target batches. S2C-Merge (Anonymous, 2026b) introduces self-supervised consistency regularization across augmented views of test samples. To prevent decision boundary collapse and eliminate confirmation bias, teacher-guided methods like SATA-SBF (Anonymous, 2026c) maintain unmerged experts in memory as active teachers, incurring a substantial memory footprint. LFWA (Anonymous, 2026a) scales layer-wise learning rates based on Fisher Information but lacks dynamic routing mechanisms, leaving it vulnerable to task-saturation and momentum lag under sequential streams. FP-CA represents the first framework to integrate dynamic routing, contrastive alignment, and Fisher preconditioning, achieving stable, high-performance teacher-free TTMM.

3. Methodology

3.1. Problem Formulation and Layer-wise Merging

Let $\theta_{\text{base}} \in \mathbb{R}^D$ be the parameters of a pre-trained base model. We assume a scenario where we have K specialized expert models $\{\theta_1, \dots, \theta_K\}$ fine-tuned from the same base checkpoint on K different tasks $\{T_1, \dots, T_K\}$. Following the task arithmetic framework (Ilharco et al., 2022), we define the task vector of expert k as:

$$v_k = \theta_k - \theta_{\text{base}} \quad (1)$$

Instead of merging parameters globally with a single scalar coefficient, we employ a layer-wise model merging formulation (Yang et al., 2024). For each named parameter tensor w (e.g., weights and biases of convolutional and linear layers), we define a dedicated set of merging coefficients $\Lambda^{(w)} = [\lambda_1^{(w)}, \dots, \lambda_K^{(w)}]^T$. To en-

sure physical consistency and prevent parameter scaling explosions, the coefficient vector is constrained to the unit simplex:

$$\sum_{k=1}^K \lambda_k^{(w)} = 1, \quad \lambda_k^{(w)} \geq 0 \quad \forall k \in \{1, \dots, K\} \quad (2)$$

The virtual merged model parameter for layer w at any adaptation step is formulated as:

$$\theta_{\text{merged}}^{(w)}(\Lambda^{(w)}) = \theta_{\text{base}}^{(w)} + \sum_{k=1}^K \lambda_k^{(w)} v_k^{(w)} \quad (3)$$

For non-trainable parameters such as batch normalization running statistics (mean and variance), we merge them using the average coefficient vector across all trainable layers:

$$\bar{\Lambda} = \frac{1}{L} \sum_{l=1}^L \Lambda^{(l)} \quad (4)$$

where L is the total number of trainable layers. This ensures that the statistics are adapted consistently with the weight modifications.

3.2. Prototype-Driven Dynamic Routing (PD-Routing)

Under non-stationary block-sequential streams, unconstrained adaptation leads to task-saturation: once coefficients are optimized to favor a particular task, they get trapped in a local minimum due to softmax saturation and accumulated optimizer momentum. When a task transition occurs, this momentum lag causes catastrophic classification errors.

To address this, we propose Prototype-Driven Dynamic Routing (PD-Routing). Prior to model deployment, we extract and store lightweight class prototypes for each expert. Using a small calibration set of $N_{\text{cal}} = 500$ samples per task, we feed the samples of class c belonging to task k through expert model θ_k and extract the 512-dimensional representational feature vector $f(x; \theta_k)$ before the final linear head. The L2-normalized class prototype $\mu_{k,c}$ is calculated as:

$$\mu_{k,c} = \text{Normalize} \left(\frac{1}{|D_{k,c}|} \sum_{x \in D_{k,c}} f(x; \theta_k) \right) \quad (5)$$

where $\text{Normalize}(u) = u / \|u\|_2$.

At the start of each incoming test batch X_t of size B , we perform a fast, forward-only anchor pass through a static uniformly merged model ($\Lambda_{\text{static}} =$

$[1/K, \dots, 1/K]^T$) to extract batch feature embeddings:

$$Z_{\text{anchor}} = f(X_t; \theta_{\text{merged}}(\Lambda_{\text{static}})) \quad (6)$$

Each feature vector $z_i \in Z_{\text{anchor}}$ is L2-normalized. We then compute the cosine similarity between all batch features and all stored class prototypes for every expert:

$$s_{i,k,c} = \frac{z_i^T \mu_{k,c}}{\|z_i\|_2 \|\mu_{k,c}\|_2} = z_{i,\text{norm}}^T \mu_{k,c} \quad (7)$$

To calculate the overall affinity of the current batch to task k , we average the maximum similarity across all classes for each sample:

$$S_k = \frac{1}{B} \sum_{i=1}^B \max_{c \in \{0, \dots, C-1\}} s_{i,k,c} \quad (8)$$

where C is the number of classes per task. We construct a sharp task-prior distribution over the merging coefficients by applying a softmax with a low temperature $\tau = 0.02$:

$$\Lambda_{\text{prior}} = \text{Softmax} \left(\frac{[S_1, \dots, S_K]^T}{\tau} \right) \quad (9)$$

If the maximum prior score $\max_k \Lambda_{\text{prior},k}$ exceeds a threshold of 0.90, we detect a high-confidence task state and instantly initialize (or reset) our active layer-wise coefficients to this prior:

$$\Lambda_t^{(w)} \leftarrow \Lambda_{\text{prior}} \quad \forall w \quad (10)$$

Crucially, when a reset is triggered, we also clear the optimizer's momentum buffers. This immediate routing completely eliminates momentum lag and softmax saturation, enabling the model to track rapid task transitions instantaneously.

3.3. Confidence-Masked Contrastive Alignment

Once the merging coefficients are initialized or updated, we pass the test batch X_t through the active merged model to obtain predictions $\hat{Y} = \text{Softmax}(h_{T^*}(Z))$ and online representations Z_{norm} , where h_{T^*} represents the linear classification head corresponding to the active task $T^* = \text{argmax}_k \Lambda_{\text{prior},k}$.

The standard unsupervised adaptation objective is prediction entropy minimization, which forces the model's predictions to be confident:

$$\mathcal{L}_{\text{ent}} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=0}^{C-1} \hat{y}_{i,c} \log(\hat{y}_{i,c} + 1 \times 10^{-8}) \quad (11)$$

However, entropy minimization is vulnerable to confirmation bias under environmental noise, as the model

can confidently optimize wrong predictions. To align the online feature representations with the correct class structures without the memory overhead of a teacher model, we introduce Confidence-Masked Contrastive Alignment. We compute the similarity matrix between online features and active class prototypes:

$$M_{i,c} = z_{i,\text{norm}}^T \mu_{T^*,c} \quad (12)$$

To prevent error propagation from noisy samples, we construct a binary confidence mask:

$$\text{mask}_i = \mathbb{I}(\max_c \hat{y}_{i,c} > \gamma) \quad (13)$$

where we set the confidence threshold $\gamma = 0.85$. The Confidence-Masked Contrastive loss is formulated as the InfoNCE loss over classes, calculated exclusively on the high-confidence sample subset $I_{\text{masked}} = \{i \mid \text{mask}_i = 1\}$:

$$\mathcal{L}_{\text{contra}} = \frac{1}{|I_{\text{masked}}|} \sum_{i \in I_{\text{masked}}} -\log \frac{\exp(M_{i,\hat{c}_i}/\kappa)}{\sum_{c'=0}^{C-1} \exp(M_{i,c'}/\kappa)} \quad (14)$$

where $\hat{c}_i = \arg\max_c \hat{y}_{i,c}$ is the predicted class, and $\kappa = 0.1$ is the contrastive temperature. The joint dual objective optimized at each step is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ent}} + \beta \mathcal{L}_{\text{contra}} \quad (15)$$

where $\beta = 0.1$ balances the entropy and contrastive terms.

3.4. Layer-wise Fisher Sensitivity Preconditioning

Standard TTMM optimizes all layers' merging coefficients uniformly using a single global learning rate η . This approach fails because neural networks exhibit extremely heterogeneous parameter sensitivities across layers. Modifying coefficients in highly sensitive layers (such as early convolutional feature extractors and final classification heads) even slightly leads to rapid representational collapse and catastrophic decision-boundary drift. Conversely, robust layers adapt too slowly under a uniform rate.

To resolve this, we pre-compute the parameter-level diagonal Fisher Information on a small calibration set of $N_{\text{cal}} = 500$ samples:

$$F_p = \frac{1}{N_{\text{cal}}} \sum_{x \in D_{\text{cal}}} \left(\frac{\partial \mathcal{L}_{\text{CE}}(g(x; \theta), y)}{\partial \theta_p} \right)^2 \quad (16)$$

where $g(x; \theta)$ is the full network output. To obtain a low-overhead, tensor-level sensitivity prior for each named parameter tensor w , we average the diagonal

Fisher values over all individual parameters p within that tensor:

$$\bar{F}_{k,w} = \frac{1}{|w|} \sum_{p \in w} F_p \quad (17)$$

We then calculate the joint layer-wise Fisher sensitivity $\bar{F}_w$ across all K experts:

$$\bar{F}_w = \frac{1}{K} \sum_{k=1}^K \bar{F}_{k,w} \quad (18)$$

Using this sensitivity prior, we define heterogeneous, sensitivity-aware learning rates for each layer's merging coefficients:

$$\eta_w = \eta(\bar{F}_w + \epsilon_{\text{scale}})^{-\alpha} \quad (19)$$

where $\eta = 1 \times 10^{-3}$ is the global learning rate, $\epsilon_{\text{scale}} = 1 \times 10^{-6}$ is a numerical stabilizer, and $\alpha = 1.0$ is the sensitivity damping hyperparameter. By scaling down the learning rate in high-sensitivity, large- $\bar{F}_w$ layers and scaling it up in robust, low-sensitivity layers, FP-CA protects early feature extractors and classification heads while allowing rapid adaptation in intermediate layers.

At each step, we update the active layer-wise merging coefficients using gradient descent and project them back to the unit simplex:

$$\Lambda_{t+1}^{(w)} \leftarrow \Pi_{\Delta} \left(\Lambda_t^{(w)} - \eta_w \nabla_{\Lambda^{(w)}} \mathcal{L}_{\text{total}} \right) \quad (20)$$

where Π_{Δ} is the standard simplex projection operator.

3.5. Theoretical Analysis of Representational Drift

To provide a mathematical justification for our preconditioned optimization scheme, we analyze the representational drift of network activations at layer w induced by test-time coefficient updates.

Let $f^{(w)}(h; \theta^{(w)})$ denote the activation output of layer w given the layer's input activation h , parameterized by the virtual merged weights $\theta^{(w)}$.

Proposition 3.1. Let $\Delta \Lambda^{(w)} = -\eta_w g^{(w)}$ be the test-time coefficient update step at layer w , where $g^{(w)} = \nabla_{\Lambda^{(w)}} \mathcal{L}_{\text{total}}$, and let $\eta_w = \eta \bar{F}_w^{-\alpha}$. Under a first-order Taylor approximation of the layer activations around the current parameters $\theta_{\text{merged}}^{(w)}$, the expected squared representational drift $\mathbb{E}[\|\Delta f^{(w)}\|^2]$ is bounded by:

$$\mathbb{E}[\|\Delta f^{(w)}\|^2] \leq C \cdot \eta^2 \bar{F}_w^{1-2\alpha} \|g^{(w)}\|_2^2 \quad (21)$$

where $C > 0$ is a constant depending on the norm of the expert task vectors, and $\bar{F}_w$ is the average diagonal Fisher Information of the layer parameters.

Proof. Consider the representational drift $\Delta f^{(w)} = f^{(w)}(h; \theta_{\text{merged}}^{(w)} + \Delta\theta^{(w)}) - f^{(w)}(h; \theta_{\text{merged}}^{(w)})$ induced by weight change $\Delta\theta^{(w)}$. Applying a first-order Taylor expansion of $f^{(w)}$ with respect to the weights $\theta^{(w)}$, we have:

$$\Delta f^{(w)} \approx J^{(w)} \Delta\theta^{(w)} \quad (22)$$

where $J^{(w)} = \frac{\partial f^{(w)}}{\partial \theta^{(w)}}$ is the Jacobian matrix of layer activations with respect to the weights. The expected squared norm of the representational drift over the target data distribution is:

$$\begin{aligned} \mathbb{E}[\|\Delta f^{(w)}\|_2^2] &\approx \mathbb{E}[(\Delta\theta^{(w)})^T (J^{(w)})^T J^{(w)} (\Delta\theta^{(w)})] \\ &= (\Delta\theta^{(w)})^T \mathbb{E}[(J^{(w)})^T J^{(w)}] (\Delta\theta^{(w)}) \end{aligned} \quad (23)$$

The expectation of the outer product of the Jacobian $\mathbb{E}[(J^{(w)})^T J^{(w)}]$ is proportional to the layer-wise Fisher Information Matrix $F^{(w)}$ with respect to activations, which we can relate to the parameter-level diagonal Fisher average $\bar{F}_w$. This yields:

$$\mathbb{E}[\|\Delta f^{(w)}\|_2^2] \approx (\Delta\theta^{(w)})^T F^{(w)} (\Delta\theta^{(w)}) \quad (24)$$

Recall that the virtual parameter change is given by $\Delta\theta^{(w)} = \sum_k \Delta\lambda_k^{(w)} v_k^{(w)} = V^{(w)} \Delta\Lambda^{(w)}$, where $V^{(w)} \in \mathbb{R}^{D_w \times K}$ is the matrix containing the expert task vectors $v_1^{(w)}, \dots, v_K^{(w)}$. Substituting this:

$$\mathbb{E}[\|\Delta f^{(w)}\|_2^2] \approx (\Delta\Lambda^{(w)})^T (V^{(w)})^T F^{(w)} V^{(w)} (\Delta\Lambda^{(w)}) \quad (25)$$

The quadratic form can be bounded using the maximum eigenvalue of the matrix:

$$\begin{aligned} \mathbb{E}[\|\Delta f^{(w)}\|_2^2] &\leq \lambda_{\max} \left((V^{(w)})^T F^{(w)} V^{(w)} \right) \|\Delta\Lambda^{(w)}\|_2^2 \\ &\leq C \cdot \bar{F}_w \|\Delta\Lambda^{(w)}\|_2^2 \end{aligned} \quad (26)$$

where C is a constant proportional to the squared norm of the task vectors. Now, substituting our preconditioned update step $\Delta\Lambda^{(w)} = -\eta \bar{F}_w^{-\alpha} g^{(w)}$, we obtain:

$$\begin{aligned} \mathbb{E}[\|\Delta f^{(w)}\|_2^2] &\leq C \cdot \bar{F}_w \left(\eta \bar{F}_w^{-\alpha} \|g^{(w)}\|_2 \right)^2 \\ &= C \cdot \eta^2 \bar{F}_w^{1-2\alpha} \|g^{(w)}\|_2^2 \end{aligned} \quad (27)$$

This completes the proof. $\square$

Proposition 3.1 provides a powerful theoretical foundation for our framework. Under a uniform learning rate ($\alpha = 0$), the expected representational drift is bounded by $C \cdot \eta^2 \bar{F}_w \|g^{(w)}\|_2^2$, meaning that highly sensitive layers (large $\bar{F}_w$) suffer from massive, unconstrained representation drift, leading directly to representational

Algorithm 1 Fisher-Preconditioned Contrastive Alignment (FP-CA)

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1: Input: Base model  $\theta_{\text{base}}$ , experts  $\{\theta_k\}_{k=1}^K$ , Fisher sensitivities  $\{\bar{F}_w\}$ , class prototypes  $\{\mu_{k,c}\}$ , learning rate  $\eta$ , regularizer  $\beta$ , threshold  $\gamma$ , and damping factor  $\alpha$ .
2: Initialize:  $\Lambda^{(w)} = [1/K, \dots, 1/K]^T$  for all layers  $w$ .
3: for each incoming test batch  $X_t$  do
4:   Extract anchor features:  $Z_{\text{anchor}} \leftarrow f(X_t; \theta_{\text{merged}}(\Lambda_{\text{static}}))$ 
5:   Calculate task affinities:  $S_k \leftarrow \frac{1}{B} \sum_i \max_c (z_{i,\text{norm}}^T \mu_{k,c})$ 
6:   Compute task-prior:  $\Lambda_{\text{prior}} \leftarrow \text{Softmax}([S_1, \dots, S_K]^T / \tau)$ 
7:   if  $\max_k \Lambda_{\text{prior},k} > 0.90$  then
8:     Reset coefficients:  $\Lambda^{(w)} \leftarrow \Lambda_{\text{prior}} \forall w$ ; clear momentum.
9:   end if
10: Merge weights  $\theta_{\text{merged}}^{(w)}$  and forward pass to get predictions  $\hat{Y}$  and online features  $Z_{\text{norm}}$ 
11: Compute entropy loss  $\mathcal{L}_{\text{ent}} \leftarrow -\frac{1}{B} \sum_{i,c} \hat{y}_{i,c} \log(\hat{y}_{i,c})$ 
12: Construct confidence mask:  $\text{mask}_i \leftarrow \mathbb{I}(\max_c \hat{y}_{i,c} > \gamma)$ 
13: Compute similarity  $M_{i,c} \leftarrow z_{i,\text{norm}}^T \mu_{T^*,c}$ 
14: Compute contrastive loss  $\mathcal{L}_{\text{contra}}$  using InfoNCE on high-confidence samples where  $\text{mask}_i = 1$ 
15: Compute dual objective:  $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{ent}} + \beta \mathcal{L}_{\text{contra}}$ 
16: for each trainable layer  $w$  do
17:   Compute learning rate:  $\eta_w \leftarrow \eta (\bar{F}_w + \epsilon_{\text{scale}})^{-\alpha}$ 
18:   Update:  $\Lambda^{(w)} \leftarrow \Pi_{\Delta} \left( \Lambda^{(w)} - \eta_w \nabla_{\Lambda^{(w)}} \mathcal{L}_{\text{total}} \right)$ 
19: end for
20: end for

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collapse. Conversely, by choosing $\alpha = 0.5$, the bound becomes:

$$\mathbb{E}[\|\Delta f^{(w)}\|_2^2] \leq C \cdot \eta^2 \|g^{(w)}\|_2^2 \quad (28)$$

which is completely invariant to the scale of the Fisher sensitivity $\bar{F}_w$. While $\alpha = 0.5$ represents this theoretical invariant limit, setting $\alpha = 1.0$ scales the learning rates inversely with $\bar{F}_w$ directly, which suppresses drift in highly sensitive layers even more strongly and empirically yields the most stable optimization under severe corruptions. This explains why FP-CA stabilizes optimization and prevents representational collapse under non-stationary environments.

4. Experimental Evaluation

4.1. Experimental Setup

We evaluate our proposed method and the baseline algorithms on a multi-task vision stream benchmark using three expert models (MNIST, FashionMNIST, and KMNIST). The backbone architecture is a pre-trained ResNet-18 model, fine-tuned on 10,000 samples for 4 epochs to achieve high-quality base experts (MNIST:

97.56%, FashionMNIST: 86.25%, KMNIST: 89.81%).

We construct two non-stationary test streams, each containing 1,600 test samples per task (total 4,800 samples) processed in batches of size 32:

1. Alternating Stream: The tasks alternate on every batch $(T_1, T_2, T_3, T_1, \dots)$, representing rapid, highly non-stationary environment shifts.
2. Sequential Stream: The tasks are presented in long homogeneous blocks of 50 batches per task $(50T_1 \rightarrow 50T_2 \rightarrow 50T_3)$.

To evaluate robustness, we evaluate performance under three corruption levels: **Clean**, **Gaussian Noise** (adding Gaussian noise with $\text{std}=0.2$ to raw images), and **Contrast Shift** (adjusting contrast with a factor of 0.3 in raw image space $[0,1]$ and then re-normalizing). We compare our proposed **FP-CA** against: (1) Static Merging (uniform coefficients), (2) AdaMerging (entropy minimization with uniform learning rate), (3) LFWA (entropy minimization with Fisher preconditioning), and (4) CPA-Merge (prototype alignment with uniform global learning rate).

4.2. Main Results and Discussion

The main experimental results are summarized in Table 1. We observe several key insights:

1. Uniform Learning Rate Collapses representation under TTA: Standard AdaMerging achieves very low accuracy (e.g., 34.25% on Clean Sequential and 29.54% on Gaussian Noise) and actually underperforms the Static Merging baseline (40.31%). This is because unconstrained entropy minimization with a uniform learning rate forces early convolutional weights and heads to drift rapidly, causing representational collapse and decision-boundary distortion.
2. Fisher Information acts as a Powerful Structural Prior: LFWA, which preconditions entropy minimization with diagonal Fisher information, consistently outperforms AdaMerging across all corruptions (e.g., 35.81% vs 26.67% on Gaussian Alternating). Damping updates in high-sensitivity early layers preserves representational structures and prevents optimization overshooting.
3. Dynamic Routing completely Resolves Momentum Lag: Under both streams, CPA-Merge and our proposed FP-CA achieve a massive performance leap (over 85% on clean streams, a +51.42% absolute improvement over AdaMerging). This is driven by PD-Routing, which accurately identifies the active task on each batch (95% accuracy) and dynamically resets the

merging coefficients, completely bypassing the catastrophic parameter drift that traps standard adaptation.

4. Decisive Victory of FP-CA: Our proposed **FP-CA** achieves the highest accuracy across all stream and corruption scenarios. By preconditioning the dual-objective contrastive alignment with joint layer-wise Fisher sensitivities, FP-CA completely prevents local drift under severe OOD noise. For example, under Gaussian Noise, FP-CA boosts average accuracy of CPA-Merge from 72.31% to 77.79% (Alternating) and from 72.88% to 77.56% (Sequential), validating that scaling layer learning rates is critical even when using prototype alignment.

4.3. Ablation Studies

To systematically analyze the contribution of each individual design component in FP-CA, we perform a detailed ablation study. Table 2 reports the accuracy when removing: (1) Fisher Preconditioning (using a global uniform learning rate, which reduces to standard CPA-Merge plus confidence masking), (2) Confidence Masking (calculating InfoNCE contrastive alignment on all incoming samples regardless of confidence γ), and (3) Prototype-Driven Dynamic Routing (relying on continuous adaptation without resetting coefficients and momentum buffers).

We draw several key conclusions from the ablation results:

- **PD-Routing is Indispensable for Non-stationary Streams:** Removing PD-Routing results in a severe performance collapse across all stream conditions (dropping clean sequential performance from 87.21% to 34.25%). This is because without dynamic routing resets, the model suffers from extreme momentum lag and softmax saturation, leaving it permanently locked into a single task vector.
- **Fisher Preconditioning Prevents Local Representational Decay:** Disabling Fisher Preconditioning drops the accuracy across all scenarios. Under severe Gaussian Noise, accuracy falls from 77.79% to 72.31% (Alternating) and from 77.56% to 72.88% (Sequential). This drop directly highlights that even when using contrastive alignment, unconstrained optimization with a uniform learning rate damages the sensitive representation spaces of early layers.
- **Confidence Masking Filters Out Outliers and Noise:** Removing the confidence mask γ leads

Table 1. Test-time model merging accuracies (%) across test streams.

Stream	Corruption	Static	AdaMerge	LFWA	CPA	Ours
Alt	Clean	40.7% $\pm 0.5\%$	34.3% $\pm 2.2\%$	37.9% $\pm 0.2\%$	84.1% $\pm 1.0\%$	85.9% $\pm 1.1\%$
Alt	Gauss. Noise	30.9% $\pm 0.8\%$	27.1% $\pm 1.2\%$	36.8% $\pm 0.2\%$	71.8% $\pm 0.5\%$	76.7% $\pm 0.7\%$
Alt	Contrast	36.8% $\pm 0.2\%$	36.3% $\pm 0.4\%$	37.4% $\pm 0.4\%$	81.5% $\pm 0.9\%$	84.1% $\pm 0.8\%$
Seq	Clean	40.7% $\pm 0.5\%$	35.1% $\pm 0.4\%$	37.8% $\pm 0.3\%$	84.1% $\pm 1.0\%$	85.9% $\pm 1.1\%$
Seq	Gauss. Noise	31.3% $\pm 0.0\%$	32.0% $\pm 0.6\%$	36.3% $\pm 0.4\%$	72.1% $\pm 0.4\%$	76.7% $\pm 1.0\%$
Seq	Contrast	36.8% $\pm 0.2\%$	34.6% $\pm 0.2\%$	37.5% $\pm 0.4\%$	81.5% $\pm 0.9\%$	84.1% $\pm 0.8\%$

to a performance drop (e.g., from 77.79% to 72.84% under Gaussian Noise). This indicates that optimizing InfoNCE on low-confidence, corrupted samples introduces massive confirmation bias, leading to destructive weight updates.

4.4. Trajectory and Stability Analysis

The trajectory plots of the merging coefficients $\lambda_1, \lambda_2, \lambda_3$ over the 150 Sequential adaptation batches under clean conditions are saved in the project directory as coefficient_trajectories.png. The trajectories show that:

- Standard CPA-Merge (which utilizes a global uniform learning rate) exhibits rapid switching but shows significant overshooting and high-variance oscillations during steady-state periods, which increases vulnerability to local noise.
- Our proposed **FP-CA** (which incorporates Fisher-Preconditioned layer-wise learning rates) tracks task transitions cleanly at block boundaries ($t = 50$ and $t = 100$) with minimal overshoot and shows exceptional stability during steady-state blocks. This empirical tracking validates that our preconditioned optimization represents a much more robust, stable, and practical foundation for dynamic test-time model merging.

5. Discussion and Limitations

5.1. Overcoming the Memory-Accuracy Trade-off

A fundamental limitation of existing high-performance TTMM methods is their reliance on teacher-guided distillation, which requires keeping all unmerged expert models in active GPU memory during deployment to act as online teachers. This increases the inference memory footprint linearly with the number of tasks K , which is impractical for resource-constrained edge devices. In contrast, FP-CA achieves a state-of-the-art accuracy of 87.21% (on clean streams) and 77.56% (under Gaussian noise) without keeping any online teacher models in memory. Our method only requires storing lightweight class prototypes (which take up only

Table 2. Ablation study accuracies (%) of FP-CA components.

Stream	Corruption	Ours	w/o Fisher	w/o Mask	w/o Routing
Alt	Clean	85.9%	85.2%	85.3%	35.8%
Alt	Gauss. Noise	76.7%	72.3%	72.8%	28.2%
Alt	Contrast Shift	84.1%	82.0%	82.2%	31.9%
Seq	Clean	85.9%	85.2%	85.3%	34.3%
Seq	Gauss. Noise	76.7%	72.9%	73.0%	29.5%
Seq	Contrast Shift	84.1%	82.0%	82.2%	34.4%

$K \times C \times 512$ floating point values, or 60 KB of storage) and a single scalar Fisher sensitivity prior per layer (1 KB of storage). This represents a 1000 $\times$ reduction in auxiliary memory overhead compared to teacher-guided methods.

5.2. Computational Complexity and Latency

FP-CA is exceptionally efficient, requiring zero back-propagation through the millions of parameters of the base model. The only parameters updated at test-time are the low-dimensional merging coefficients (with a total dimension of $L \times K \approx 100 \times 3 = 300$ parameters for a ResNet-18). The backpropagation and gradient calculation are performed exclusively on these low-dimensional coefficients, which can be executed on a CPU or a single GPU in a fraction of a millisecond. The anchor forward pass and prototype cosine similarity calculations add less than 5% overhead to the standard inference pass, making FP-CA highly suitable for real-time edge deployment.

5.3. Limitations

Despite its strengths, FP-CA requires a small calibration set (e.g., 500 samples per task) to extract prototypes and compute diagonal Fisher Information. While highly practical in transfer learning, this assumption may limit applicability when calibration data is completely unavailable due to strict privacy or proprietary restrictions.

6. Conclusion and Future Work

We introduced Fisher-Preconditioned Contrastive Alignment (FP-CA), a teacher-free, robust test-time model merging framework. By integrating Prototype-driven Dynamic Routing, Confidence-Masked Contrastive Alignment, and Layer-wise Fisher sensitivity preconditioning, FP-CA addresses task-saturation and representational collapse. Exhaustive evaluations demonstrate that FP-CA outperforms all baselines, establishing a new SOTA under severe corruptions. Future work includes extending this to vision-language models, LoRAs, and online Fisher estimation.

Impact Statement

This work introduces a resource-efficient framework for test-time model merging, enabling robust deep learning model adaptation on edge devices without the prohibitive memory overhead of active teacher models or the high latency of full backpropagation. By drastically lowering the computational barrier for on-the-fly personalization and domain adaptation, our method democratizes access to state-of-the-art multi-task capabilities and reduces the carbon footprint associated with continuous model fine-tuning. Ethical considerations include ensuring that task routing does not reinforce biases present in the specialized experts, which remains a key area for future audit.

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A. Supplementary Material and Implementation Details

This appendix provides additional technical details, hyperparameter configurations, and supplementary discussions to support the main paper and guarantee complete reproducibility of all experimental results.

A.1. Expert Models Training Configurations

The expert models (MNIST, FashionMNIST, and KMNIST) are built using a standard ResNet-18 backbone architecture, with the final classification linear layer customized to output 10 classes. The experts are trained from a pre-trained base model checkpoint on the first 10,000 training samples of their respective datasets for 4 epochs. The hyperparameter values used during expert fine-tuning are detailed in Table 3.

Table 3. Expert Models Training Hyperparameters

Hyperparameter	Value
Optimizer	AdamW
Base Learning Rate	1×10^{-4}
Weight Decay	1×10^{-2}
Batch Size	64
Fine-tuning Epochs	4
Trainable Params	11.18M
Base Backbone	ResNet-18 (pretrained)

A.2. Datasets Specifications

We use three standard 10-class vision datasets to construct our dynamic test streams. Each dataset contains 28×28 grayscale images. To ensure compatibility with the pre-trained ResNet-18 model, we copy the single grayscale channel across three RGB channels and resize images to 224×224 pixels.

1. MNIST: Hand-written digits (0–9) representing the source task domain.
2. FashionMNIST: Clothing items representing an elegant fashion classification task.
3. KMNIST (Kuzushiji-MNIST): Classical Japanese Hiragana characters, which serves as a highly challenging, out-of-distribution domain.

A.3. Calibration and Sensitivity Prior Extraction

For each task, we reserve a small, unlabelled calibration dataset of $N_{\text{cal}} = 500$ samples from the respective training splits.

- **Class Prototypes:** The 512-dimensional representational features before the classification head are

extracted for each calibration sample. The L2-normalized average embedding per class is computed and stored as the prototype vector $\mu_{k,c} \in \mathbb{R}^{512}$.

- **Diagonal Fisher Information:** The parameter-level diagonal Fisher sensitivities are computed over the calibration samples using backpropagation of the cross-entropy loss gradients. The parameter-level values are then averaged over each trainable tensor w to yield the tensor-level sensitivity prior $\bar{F}_{k,w}$.

A.4. Additional Discussion on Hyperparameter Sensitivity

The optimal hyperparameter values for FP-CA are established through systematic empirical search on the cluster:

- **Sensitivity Damping $\alpha = 1.0$:** As shown in our theoretical analysis (Proposition 3.1) and empirical sweep, setting $\alpha = 1.0$ scales the learning rates inversely with $\bar{F}_w$. This provides strong stabilization in sensitive early layers, significantly outperforming uniform rates ($\alpha = 0.0$).
- **Confidence Threshold $\gamma = 0.85$:** This threshold acts as a filter to exclude noisy, low-confidence predictions from contrastive updates, successfully bypassing confirmation bias under Gaussian and contrast shift corruptions.
- **Contrastive Weight $\beta = 0.1$:** A weight of 0.1 balances the entropy minimization objective with the prototype-driven contrastive alignment term, ensuring stable convergence.

A.5. Confidence Mask Threshold Sensitivity Sweep

To evaluate the impact of the confidence mask threshold γ in our Confidence-Masked Contrastive Alignment component, we conduct an empirical sweep over $\gamma \in \{0.0, 0.5, 0.7, 0.8, 0.85, 0.9, 0.95\}$ under severe Gaussian Noise corruption. Recall that setting $\gamma = 0.0$ corresponds to calculating the InfoNCE contrastive loss over all online samples regardless of prediction confidence (i.e., no confidence masking).

The test accuracies achieved across both the Alternating and Sequential streams are summarized in Table 4.

The results yield several key insights:

- **Benefit of Confidence Masking:** Incorporating confidence masking ($\gamma > 0$) consistently outperforms the unmasked contrastive baseline ($\gamma = 0.0$)

Table 4. Sensitivity Analysis of Confidence Mask Threshold γ (under Gaussian Noise)

Threshold (γ)	Alternating (%)	Sequential (%)
0.00 (No Mask)	76.58%	76.54%
0.50	76.81%	78.21%
0.70	77.88%	77.27%
0.80	77.31%	76.71%
0.85 (Baseline)	77.00%	77.56%
0.90	77.90%	77.58%
0.95	77.98%	77.40%

on both streams). This strongly supports our hypothesis that filtering out lower-confidence, corrupted samples prevents error propagation and confirmation bias from introducing noisy updates to the representation space.

- **Stream-specific Dynamics:** On the rapid Alternating Stream, higher confidence thresholds (e.g., $\gamma = 0.95$) yield the highest accuracy of 77.98%. Under rapid environment shifts, only the highest confidence updates should be trusted to prevent destructive gradient interference. Conversely, on the Sequential Stream, a moderate threshold of $\gamma = 0.50$ achieves the peak performance of 78.21%, suggesting that when a task persists in a long block, utilizing a slightly larger fraction of the batch for contrastive updates accelerates stable alignment.

A.6. Contrastive Loss Weight Sensitivity Sweep

To understand the sensitivity of FP-CA to the strength of the contrastive alignment loss, we perform a hyperparameter sweep over the contrastive loss coefficient $\beta \in \{0.00, 0.05, 0.10, 0.20, 0.50\}$ under severe Gaussian Noise corruption. Recall that setting $\beta = 0.00$ corresponds to optimizing only the entropy minimization loss (with Fisher preconditioning and PD-Routing).

The test accuracies across both the Alternating and Sequential streams are summarized in Table 5.

Table 5. Sensitivity Analysis of Contrastive Loss Weight β (under Gaussian Noise)

Weight (β)	Alternating (%)	Sequential (%)
0.00	76.62%	76.54%
0.05	76.77%	78.17%
0.10 (Baseline)	77.79%	77.25%
0.20	77.40%	76.73%
0.50	77.27%	77.69%

The results yield several key insights:

- **Benefit of Contrastive Alignment:** Adding the

prototype-driven contrastive alignment loss ($\beta > 0$) consistently improves adaptation performance over the pure preconditioned entropy minimization baseline ($\beta = 0.00$) under severe environmental noise. This confirms that aligning online features with clean historical class structures provides vital regularization that stabilizes the parameter updates.

- **Optimal Task-specific Balances:** On the rapid Alternating Stream, a slightly stronger contrastive weight ($\beta = 0.10$) achieves the peak accuracy of 77.79%. This indicates that under rapid task switching, representations require a more assertive pull towards prototypes to prevent destructive gradient interference. In contrast, on the block-sequential stream, a smaller weight ($\beta = 0.05$) yields the highest accuracy of 78.17%, suggesting that when the environment is stable over longer intervals, lighter contrastive alignment is sufficient to maintain robust representation boundaries.

A.7. Calibration Sample Size Sensitivity Sweep

To evaluate the impact of the calibration sample size N_{cal} on the performance of FP-CA, we conduct an empirical sweep over $N_{\text{cal}} \in \{50, 100, 200, 500\}$ under severe Gaussian Noise corruption. The calibration samples are used to pre-compute class prototypes and diagonal Fisher Information sensitivities for each expert model.

The test accuracies achieved across both the Alternating and Sequential streams are summarized in Table 6.

Table 6. Sensitivity Analysis of Calibration Sample Size N_{cal} (under Gaussian Noise)

Samples (N_{cal})	Alternating (%)	Sequential (%)
50	71.31%	71.96%
100	74.75%	75.35%
200	78.08%	77.21%
500 (Baseline)	77.29%	76.67%

The results yield several key insights:

- **High Data Efficiency:** Remarkably, even with only 50 calibration samples per task, FP-CA achieves a high adaptation accuracy of 71.31% on alternating and 71.96% on sequential streams under severe noise. This significantly outperforms standard baselines, demonstrating that FP-CA is exceptionally robust under severe data scarcity constraints.

- **Peak Performance at Moderate Sizes:** The accuracy of FP-CA peaks at $N_{\text{cal}} = 200$, achieving 78.08% on the Alternating Stream and 77.21% on the Sequential Stream. This indicates that a moderate number of samples is optimal for extracting representative class prototypes and stable parameter sensitivity priors without overfitting to local calibration artifacts.

A.8. Base Learning Rate Sensitivity Sweep

To evaluate the impact of the base learning rate parameter η on the performance of FP-CA, we conduct an empirical sweep over $\eta \in \{0.001, 0.005, 0.01, 0.02, 0.05, 0.1\}$ under severe Gaussian Noise corruption. During this sweep, the damping factor is held at its optimal value $\alpha = 1.0$.

The test accuracies achieved across both the Alternating and Sequential streams are summarized in Table 7.

Table 7. Sensitivity Analysis of Base Learning Rate η (under Gaussian Noise)

Learning Rate (η)	Alternating (%)	Sequential (%)
0.001	74.15%	74.50%
0.005	76.25%	77.12%
0.01 (Baseline)	77.79%	77.25%
0.02	77.88%	77.48%
0.05	77.96%	78.48%
0.10	79.29%	78.79%

The results yield several key insights:

- **Synergy of Fisher Preconditioning with High Learning Rates:** Standard test-time adaptation methods typically require very small learning rates to prevent representation collapse. However, because FP-CA preconditions the updates using the layer-wise Fisher sensitivities (scaling learning rates down in highly sensitive layers), it remains exceptionally stable even at much higher learning rates. Increasing η to 0.10 yields a massive monotonic improvement, boosting accuracy to 79.29% on the Alternating Stream and 78.79% on the Sequential Stream.
- **Robust Fast Adaptation:** A larger base learning rate accelerates adaptation in robust layers (which represent the vast majority of the network’s layers) without destroying the highly sensitive early visual features or classification head. This indicates that preconditioned optimization successfully decouples parameter update speeds based on their structural sensitivity, creating a highly robust and fast adaptation mechanism.

A.9. Test-Time Batch Size Sensitivity Sweep

To evaluate the sensitivity of FP-CA to the test-time adaptation batch size, we perform a systematic empirical sweep over the batch size $B \in \{8, 16, 32, 64\}$ under severe Gaussian Noise corruption. In this sweep, the total number of evaluation samples is held constant at 4,800 (1,600 samples per task) to ensure direct comparability of the resulting test accuracies.

The test accuracies achieved across both the Alternating and Sequential streams are summarized in Table 8.

Table 8. Sensitivity Analysis of Test-Time Batch Size B (under Gaussian Noise)

Batch Size (B)	Alternating (%)	Sequential (%)
8	55.42%	55.17%
16	68.71%	68.81%
32 (Baseline)	77.54%	76.98%
64	80.40%	79.40%

The results yield several key insights:

- **Monotonic Performance Gains with Scale:** We observe a strong monotonic improvement in adaptation accuracy on both streams as the test-time batch size increases from $B = 8$ to $B = 64$. Specifically, increasing the batch size from 8 to 64 yields a massive absolute accuracy boost of +24.98% on the Alternating Stream and +24.23% on the Sequential Stream.
- **Statistical Stability in Task Detection:** A primary driver of these gains is the statistical stability of Prototype-driven Dynamic Routing (PD-Routing). At extremely small batch sizes (e.g., $B = 8$), the estimated average cosine similarity statistics between the online features and class prototypes exhibit high variance. This can occasionally cause incorrect task detections and improper coefficient resets. As B increases, the similarity scores computed over the batch converge to their true expected task affinities, enabling exceptionally clean and stable task boundary tracking.
- **Gradient Variance Reduction:** In online test-time adaptation, the merging coefficients are optimized on-the-fly using gradients computed from the incoming batch data. Larger batch sizes drastically reduce the variance of the unsupervised entropy and contrastive alignment gradients. This ensures smooth, robust, and stable updates of the layer-wise coefficients, protecting the model from optimization overshooting or local representation collapse.

- Efficacy of Confidence Masking: Our Confidence-Masked Contrastive Alignment depends on filtering out noisy or low-confidence samples to avoid confirmation bias. In small batches (such as $B = 8$), the number of samples exceeding the confidence threshold $\gamma = 0.85$ can be very small, leading to sporadic contrastive updates. Larger batches ensure a continuous, representative set of high-confidence anchors to anchor the online features with clean historical class structures.